Population-proportion confidence interval

C8.01-C8.68 CORE ONLY (SKILL IDs)

1 RECOGNISE IT

- Estimate a population percentage/proportion π.
- Sample has x successes out of n.
- Need a two-sided CI and normal approximation is adequate.

2 DECISION / METHOD

1. p=x/n.
2. Check np > 5 AND n(1-p) > 5.
3. Estimated SE=√[p(1-p)/n].
4. CI=p±z*SE.
5. Convert to percent only if requested.

3 FORMULA

p=x/n
SE_hat=√[p(1-p)/n]
$z^* = \Phi^{-1}(1-\alpha/2) > 0$
CI: p ± z*SE_hat

4 FORMULA SHEET VS WHAT I KNOW

Gives: theoretical SE uses π for a sampling distribution.
Still know: a CI estimates unknown π, so substitute sample p into estimated SE.

5 EXCEL ROUTE

- p=successes/n
- SE=SQRT(p*(1-p)/n)
- z=NORM.S.INV(1-alpha/2)
- endpoints p ± z*SE.

6 TINY WORKED EXAMPLE

x=60,n=100,C=.95: p=.60,SE=.0490,margin=.0960 -> CI (.5040,.6960).

7 TOP EXAM TRAPS

- Weakening the strict sourced course inequality.
- Using null π_0 in a CI SE.
- Forgetting p is a proportion, not a percentage.
- Using π instead of sample p in the estimated CI SE.

8 SAY ALOUD

A proportion CI uses sample p both as the centre and inside the estimated standard error.

9 EXIT CHECK

Why does a CI for π use p rather than an unknown π inside its estimated SE?

ONE-GLANCE VISUAL ROUTE

